(a) Cohorts (PTB-XL, leads I and II)

Class counts: NORM / AFIB/AFL / CD / STTC/MI / HYP

Eligibility-filtered pool: Phase 3D (n = 15,160)

adults; consistent single-class assignment;
first eligible recording per patient
8,052 / 1,155 / 1,647 / 3,486 / 820

↓ quota subsample

Quota-selected cohort: Phase 3B (n = 5,000)

subset of the pool; quota seed 42;
five outer partition seeds (42–46)
1,045 / 1,045 / 1,045 / 1,045 / 820

↓ exclude 496

HYP+STTC exclusion: Phase 3C (n = 4,504)

496 co-occurring HYP+STTC patients removed;
seed 42; evaluation A (rescore), B (retrain)
1,045 / 1,045 / 1,045 / 1,045 / 324

(b) Data roles within each outer fold

Test	Student training	Val.
1,000	3,200	800

Outer test / student training / validation per fold in Phase 3B; Phase 3D uses the same proportions.

Scaler	fitted on student training only
Teacher targets	3-fold inner OOF TabPFN (4 estimators) within student training
Student targets	one-hot | 0.85·one-hot + 0.15/C | 0.5·one-hot + 0.5·teacher
Early stopping	validation macro F1, true labels (patience 15)
Teacher reference	TabPFN (8 estimators) fitted on student training
Evaluation	outer test patients only; one prediction per patient